

Attendance - Training Course on "Dubai BIM Standard & Submission Requirements"



What You Will Learn From This Workshop



The purpose of DM BIM
mandate



The BIM submission
requirements and process



How to prepare compliant
BIM submissions

Why DM BIM Mandate?

- ❑ **BIM Mandate** is a key enabler of Dubai's transition toward a fully digital construction ecosystem.
- ❑ **BIM Adoption** supports the transition toward:
 - ✓ Higher design quality and coordination
 - ✓ Advanced digital permitting and inspection services
 - ✓ Automated regulatory compliance
 - ✓ Data-driven digital twins
 - ✓ Transparency and digital integration across the lifecycle
 - ✓ AI-powered review and decision-making
 - ✓ & more ...

SOME OF THE SUBMITTED MODELS

Listed from smallest to the biggest Bim submissions



BUA: 132.11m²
File Size:16 MB



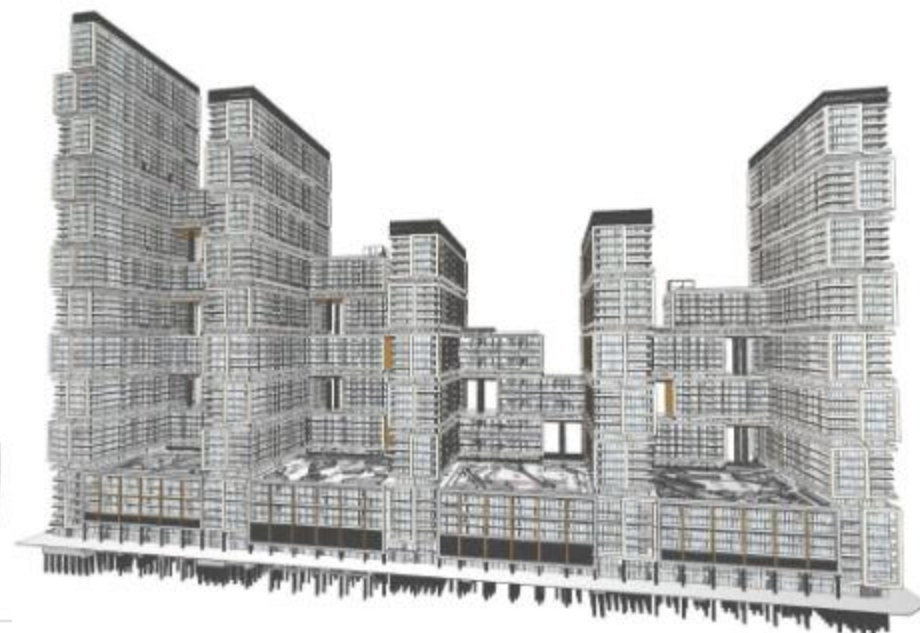
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File Size:46 MB



BUA: 28428.95 m²
File Size:126 MB



BUA: 86107.6 m²
File Size:415 MB



BUA: 553746.5 m²
File Size:2.34 GB

- BIM submissions (Jan. 2024 – April 2026) : 220
- Consultants who submitted successfully: 75



PNC ARCHITECTS

Project No. 391842



SILVER STONE ENGINEERING CONSULTANTS

Project No. 413845



FUTURE ART ENGINEERING CONSULTANCY

Project No. 414798



NEXT ENGINEERING CONSULTANTS L.L.C

Project No. 397985



ARCH GROUP CONSULTANTS

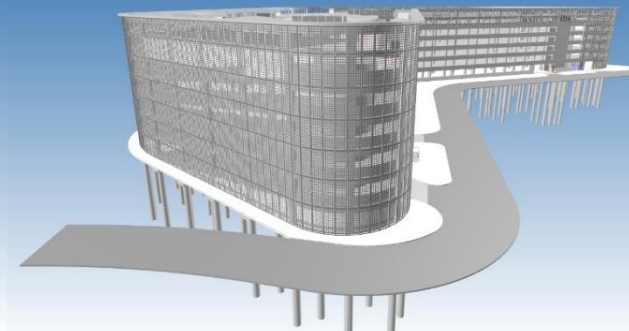
Project No. 399738



ARCH GROUP CONSULTANTS

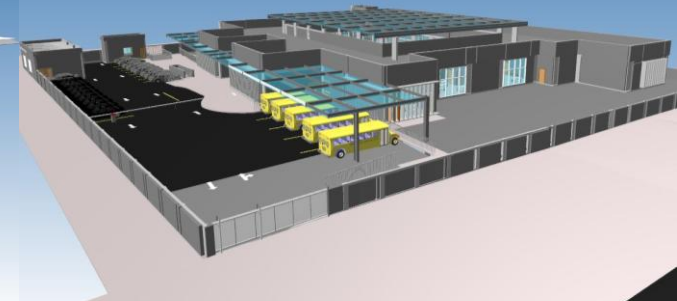
Project No. 399738

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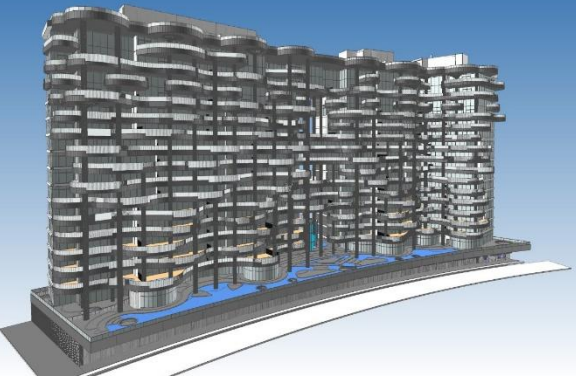
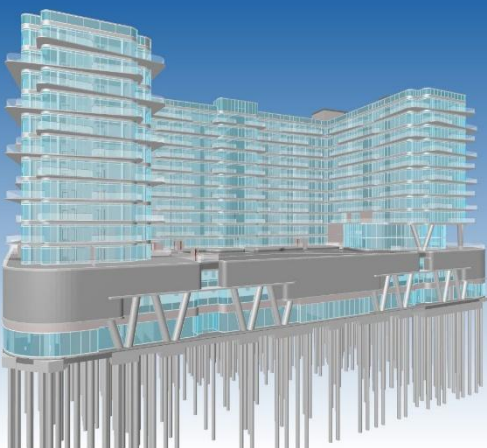
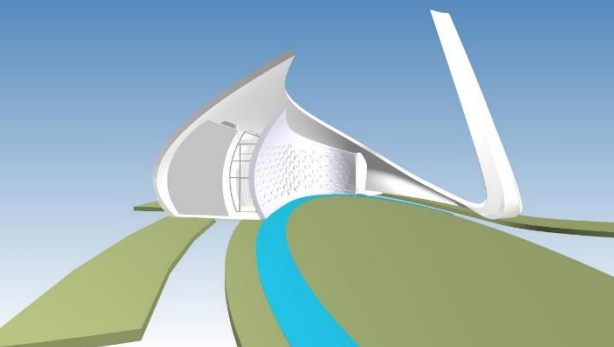
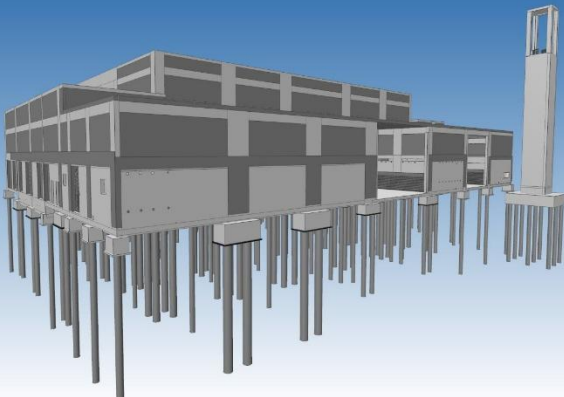
ARCH GROUP CONSULTANTS

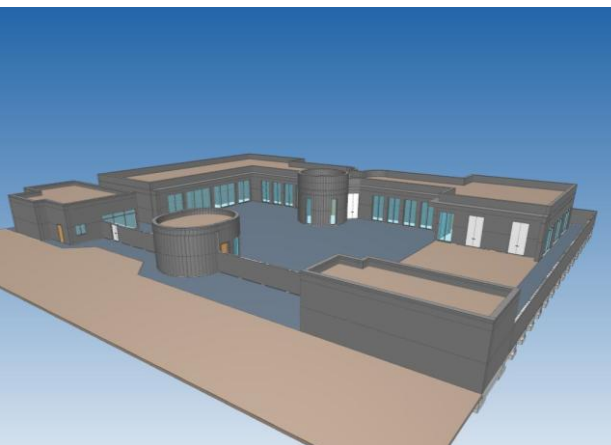
Project No. 399738



A E PRIME ENGINEERING CONSULTANTS

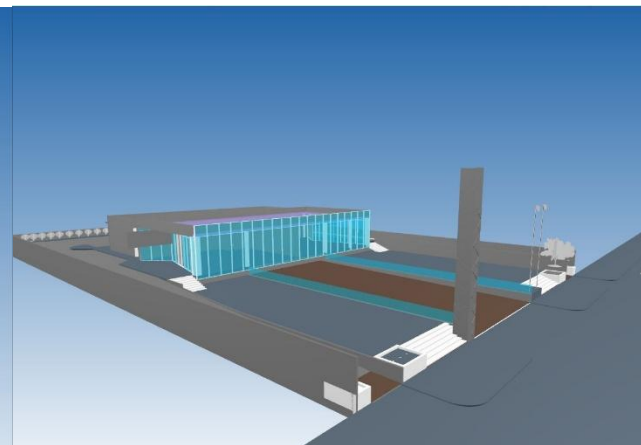
Project No. 413614

			
ALTURATH ENGINEERING CONSULTANT L.L	GULF ENGINEERING & CONSULTANTS	ENGINEERING CONSORTIUM - CONSULTING	ARKITEKNIK INTERNATIONAL CONSULTING
Project No. 353538	Project No. 384311	Project No. 408922	Project No. 410300
			
GCI	A E PRIME ENGINEERING CONSULTANTS	NAGA - ARCHITECTS & DESIGNERS & PLANNERS	ALBAIT INTERNATIONAL ARCHITECTURAL DESIGN
Project No. 416582	Project No. 418491	Project No. 422333	Project No. 424079



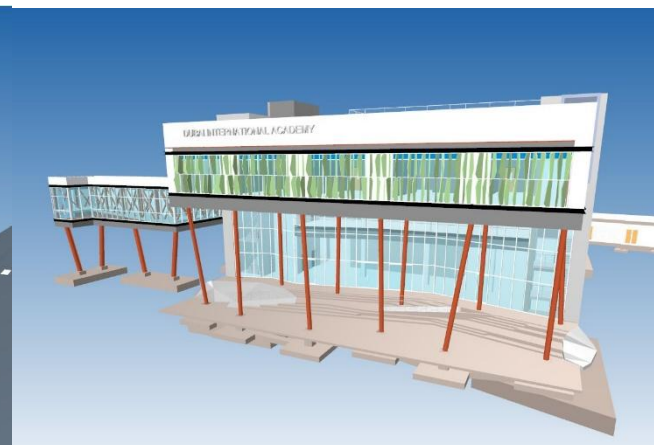
TALENT ENGINEERING CONSULTANTS L.L.C

Project No. 427309



NAD ALSHIBA ENGINEERING CONSULTANTS

Project No. 420475



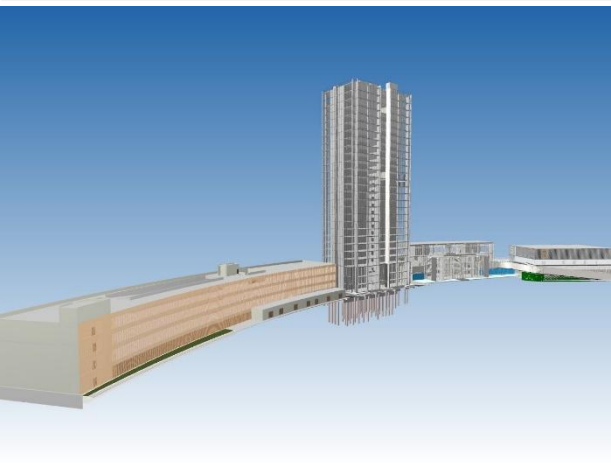
JOHN. R. HARRIS & PARTNERS

Project No. 411423



AL JABAL ENGINEERING CONSULTANCY

Project No. 402513



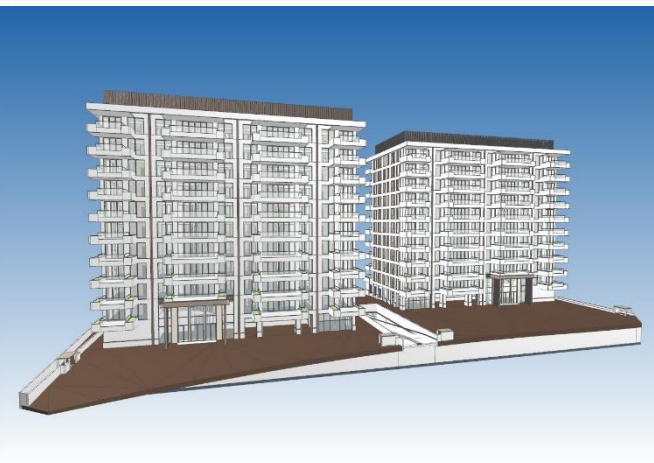
PNC ARCHITECTS

Project No. 425117



PNC ARCHITECTS

Project No. 396435



DEWAN AI EMARA CONSULTING

Project No. 412410



MAKS ARCHITECTS

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DM Mandates & Permitting Process Under the BIM Mandate

BIM Mandate (2023)

- ✓ The buildings and structures that exceed a height of 20 floors for architectural designs and 40 floors for structural designs.
- ✓ The buildings, structures, and complexes with an area exceeding 20,000 square meters for architectural designs and 30,000 square meters for structural designs.
- ✓ Specialized buildings and facilities such as hospitals and universities and the like.
- ✓ Government projects, excluding only utility infrastructure projects.




تعميم رقم (9-1-2) لسنة 2023
تاريخ 2023/10/13
بشأن إلزامية تقديم نموذج معلومات البناء (BIM Model)
مع طلب رخصة البناء لبعض أنواع المباني

السادة / مكاتب الاستشارات الهندسية وشركات مقاولات البناء المحترمين،

تحية طيبة وبعد،

في إطار جهود مؤسسة تنظيم وترخيص المباني الرامية إلى تطوير قطاع البناء والتشييد في إمارة دبي وتعزيز التحول الرقمي فيه، ولاحقاً للتعميم رقم (196) لسنة 2013 و (207) لسنة 2015 و (9-1-1) لسنة 2021، بخصوص تطبيق نمذجة معلومات البناء وما تحققة من فوائد متعددة خلال كافة مراحل دورة حياة المشروع من تقليل للتكاليف وتوفير في الوقت والجهد وتحسين كفاءة عمليات البناء والتشغيل وتقليل الاعتماد على العمالة غير الماهرة، فقد تم إطلاق خدمة التدقيق الآلي على تصاميم المباني باستخدام نمذجة معلومات البناء (BIM) وفق كود دبي للبناء ومعايير دبي القياسية لنمذجة معلومات البناء "Dubai BIM Standard" والتي يمكن الوصول لها من خلال منصة البناء في دبي (<https://buildindubai.gov.ae>).

وقد تقرر **إلزامية تقديم التصاميم المعمارية والإنشائية بهيئة (Open BIM)** وفقاً لمعايير دبي القياسية عند التقديم على رخص البناء الجديدة للمباني والمنشآت التالية كمرحلة أولى ابتداء من **2024/01/01**:

- 1- المباني والمنشآت التي يزيد ارتفاعها عن (20) طابقاً للتصاميم المعمارية وعن (40) طابقاً للتصاميم الإنشائية.
- 2- المباني والمنشآت والمجمعات التي تزيد مساحتها عن (20) ألف متر مربع للتصاميم المعمارية وعن (30) ألف متر مربع للتصاميم الإنشائية.
- 3- المباني والمنشآت التخصصية كالمستشفيات والجامعات وما في حكمها.
- 4- المشاريع الحكومية باستثناء مشاريع المرافق الخدمية فقط.

ويكون تقديم نماذج معلومات البناء خلال هذه المرحلة متوازياً مع بقية المخططات والمتطلبات التي تقدم حالياً مع رخصة البناء ولا يلغىها. كما ونوه إلى أنه لا مانع من تقديم نماذج معلومات البناء للتصاميم المعمارية للفلل الخاصة والاستثمارية والمباني غير المشمولة في هذا التعميم، حيث سيساهم ذلك في دعم تقييمات التفوق الهندسي للمكاتب الاستشارية الملتزمة بالتطبيق.

ولمزيد من المعلومات يرجى زيارة المنصة على الرابط المذكور أعلاه، والاطلاع على مستندات المعايير والمصادر التعليمية المتوفرة، كما ويمكن إرسال أية استفسارات على البريد الإلكتروني GeoDubai@dm.gov.ae.

هذا ما لزم تعميمه للعمل بما ورد فيه لتحقيق المصلحة العامة.

DM BIM Mandate (2026 Update)

What's new?

The BIM mandate now applies to:

A. New buildings:

- ✓ Buildings, facilities, and complexes higher than G+12 floors
- ✓ Buildings, facilities, and complexes with area $\geq 15,000 \text{ m}^2$

B. Additions & major modifications to existing BIM-permitted buildings

- ✓ Adding or removing floors
- ✓ Increasing area by $1,500 \text{ m}^2$ or more
- ✓ complete design changes
- ✓ Change of main building use
- ✓ Change of building location within the plot



حكومة دبي
GOVERNMENT OF DUBAI



بلدية دبي
Dubai Municipality

تعميم رقم (9-1-3) لسنة 2026
بتاريخ 2026/01/15

بشأن التوسع في الزامية تقديم نماذج معلومات البناء (BIM Models)
مع طلب رخصة البناء لبعض أنواع المباني

السادة/ مكاتب الاستشارات الهندسية وشركات مقاولات البناء المحترمين،
تحية طيبة وبعد،

في إطار جهود مؤسسة تنظيم وترخيص المباني الرامية إلى تطوير قطاع البناء والتشييد في إمارة دبي، وتعزيز التحول الرقمي وتحقيق الكفاءة التشغيلية لجميع مراحل المشروع، واستناداً إلى التعميم رقم (9-1-2) لسنة 2023 بشأن إلزامية تقديم نماذج معلومات البناء (BIM) لبعض أنواع المباني، نفيدهم بأنه قد تقرر التوسع في نطاق التطبيق، وذلك اعتباراً من 1 أبريل 2026، ليشمل ما يلي:

أولاً: المباني الجديدة:

يشمل نطاق التطبيق في هذه المرحلة الفئات التالية من المباني، وذلك فيما يخص التصميم المعماري والإنشائي:

- المباني والمنشآت والمجمعات التي يزيد ارتفاعها عن (أرضي + 12 طابقاً).
- المباني والمنشآت والمجمعات بمساحة بناء (15 ألف متر مربع) أو أكثر.

ثانياً: التعديلات والإضافات على مباني مرخصة سابقاً باستخدام نظام نمذجة معلومات البناء (BIM)

1- التعديلات الرئيسية، يجب تقديم نموذج معلومات البناء (BIM) مع طلب الترخيص، وتتضمن الآتي:

- إضافة أو إلغاء طوابق
- إضافة مساحة للمبنى بمقدار 1500م مربع أو أكثر
- تغيير التصميم بالكامل
- تغيير الاستعمال الرئيسي للمبنى
- تغيير موقع المبنى على قطعة الأرض

2- التعديلات الأخرى التي لا تندرج في البند السابق يتم تقديم نموذج معلومات البناء (BIM) النهائي قبل التقدم بطلب شهادة الإنجاز على أن يعكس الأعمال المعتمدة والمنفذة فعلياً في الموقع.

علماً بأنه يجب تقديم نماذج معلومات البناء (BIM) بشكل متزامن مع المخططات المطلوبة لترخيص المباني، على أن تكون المخططات مستخرجة مباشرة من نماذج (BIM) وبما يضمن تطابقها الكامل معها.

وفي هذا الإطار وبصورة اختيارية نحث المكاتب الاستشارية على تقديم نماذج معلومات البناء للمباني غير المشمولة حالياً، حيث سيساهم ذلك في دعم تقييمات التفوق الهندسي والتهئية للتطبيق الشامل لمنظومة نمذجة المباني.

ولمزيد من المعلومات والاطلاع على مستندات المواد التعليمية المتوفرة، يرجى زيارة منصة البناء في دبي على الرابط <https://buildindubai.gov.ae/bim> كما يمكن إرسال أية استفسارات على البريد الإلكتروني: GeoDubai@dm.gov.ae

هذا ما لزم تعميمه للعمل بما ورد فيه لتحقيق المصلحة العامة.



مريم عبيد المهيري
المدير التنفيذي لمؤسسة
تنظيم وترخيص المباني

@DubaiMunicipality
f % @ o

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dm.gov.ae

مشارك - خاص / SHARED - CONFIDENTIAL

Permitting Process Under the BIM Mandate

❑ 2D Drawings → No requirement changes

❑ Drawings must be consistent with the submitted BIM models

❑ The building permit system will **automatically identify & require BIM model submission** for the following two cases:

❑ **“Final New Budlings”** :

- ✓ higher than G+12 floors
- ✓ with area $\geq 15,000 \text{ m}^2$
- ✓ governmental and specialized buildings

❑ **“Major Modifications”** on BIM-permitted buildings that includes:

- ✓ adding or removing floors
- ✓ increasing area by $1,500 \text{ m}^2$ or more
- ✓ complete design changes
- ✓ change of main building use

❑ For **“Other modifications”** the building permit system **will not require BIM model submission**, but the system **will automatically create a condition** to submit the final updated BIM model before obtaining the completion certificate.

❑ For **“Change of Building Location”** the building permit system **will not automatically detect the change**, but the submission of the updated BIM model is mandatory.

Dubai BIM Standard Overview

Dubai BIM Standard

Dubai BIM Standard defines the mandatory information requirements and exchange formats for building permit submissions, enabling:

1. consistent regulatory review,
2. automated compliance checking,
3. and GIS integration (GeoBIM).



DubaiBIMStandard.pdf



DubaiBIMTechnicalGuides.pdf



Appendix A_Model Element Matrix.xlsx



Appendix B_Model Attribute Matrix.xlsx



Appendix C_Usages Codes.xlsx



Dubai BIM Standard - Exchange Information Requirements (EIR)

DubaiBuildingPermitBIMStandards.pdf



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Technical Guides

The guides cover the following topics:

- ☐ Model Preparation
- ☐ Model Submission
- ☐ Submission Self-Assessment



Dubai BIM Standard – Model Information Requirements

Appendix B_Model Attribute Matrix.xlsx

Appendix B_Model Attribute Matrix.xlsx • Saved to this PC			
File Home Insert Draw Page Layout Formulas Data Review View Automate Developer Help			
F34			
A	B	C	D
list of categories	Equivalent IFC Class	Disciplines	Note
Project	IfcProject	ALL	Required
Building	IfcBuilding	ALL	Required
Storey	IfcBuildingStorey	ALL	Required
Beam	IfcBeam	AR, ST	Required
Building Element Proxy	IfcBuildingElementProxy	AR, ST, MEP	Not required (hidden)
Civil Element	IfcCivilElement	AR	Required
Column	IfcColumn	AR, ST	Required
Curtain Wall	IfcCurtainWall	AR	Required
Distribution Board	IfcElectricDistributionBoard	MEP	Not required (hidden)
Ducting	IfcDuctSegment	MEP	Not required (hidden)
Duct Fitting	IfcDuctFitting	MEP	Not required (hidden)
Door	IfcDoor	AR	Required
Fan	IfcFan	MEP	Not required (hidden)
Fire Suppression	IfcFireSuppressionTerminal	MEP	Not required (hidden)
Covering (Finishes)	IfcCovering	AR	Required
Floor (Slabs)	IfcSlab	ST	Required
Foundation	IfcFooting	ST	Required
Furniture	IfcFurniture	AR	Not required (hidden)
Lighting Fixture	IfcLightFixture	MEP	Not required (hidden)
Member	IfcMember	ST	Required
Parking (Spaces)	IfcSpaceType.Parking	AR	Required
Pile	IfcPile	ST	Required
Pipe	IfcPipeSegment	MEP	Not required (hidden)
Pipe Fitting	IfcPipeFitting	MEP	Not required (hidden)
Plate	IfcPlate	ST	Required
Pump	IfcPump	MEP	Not required (hidden)
Railings	IfcRailing	AR	Required
Ramp	IfcRampFlight	AR	Required
Roof	IfcRoof	AR	Required
Room (Spaces)	IfcSpace	AR	Required
Sanitary Terminal	IfcSanitaryTerminal	MEP	Not required (hidden)
Sensor	IfcSensor	MEP	Not required (hidden)
Shading Device	IfcShadingDevice	MEP	Required

Appendix C_Usages Codes.xlsx

Appendix C_Usages Codes.xlsx • Saved to this PC				
File Home Insert Draw Page Layout Formulas Data Review View Automate				
F138				
A	B	C	D	E
Code	Zone Category	Space Description	Space Category Code	Source (for internal use)
1	Amenities	Hypermarket	SC_10_10_10	DM_GIS_BDDS_v5.3
2	Amenities	Kiosk	SC_10_10_15	DM_GIS_BDDS_v5.3
3	Amenities	Salon	SC_10_15_10	DM_GIS_BDDS_v5.3
4	Amenities	Barber	SC_10_15_15	DM_GIS_BDDS_v5.3
5	Amenities	Restaurant	SC_10_20_10	DM_GIS_BDDS_v5.3
6	Amenities	Food Court	SC_10_20_15	DM_GIS_BDDS_v5.3
7	Amenities	Open Dinning Area	SC_10_20_20	DM_GIS_BDDS_v5.3
8	Amenities	Pantry	SC_10_20_25	DM_GIS_BDDS_v5.3
9	Amenities	Gym	SC_10_25_10	DM_GIS_BDDS_v5.3
10	Amenities	Exercise Hall	SC_10_25_15	DM_GIS_BDDS_v5.3
11	Amenities	Aerobic Hall	SC_10_25_20	DM_GIS_BDDS_v5.3
12	Amenities	Fitness Area	SC_10_25_25	DM_GIS_BDDS_v5.3
13	Amenities	Kids Play Room	SC_10_25_30	DM_GIS_BDDS_v5.3
14	Amenities	Kids Play Area	SC_10_25_35	DM_GIS_BDDS_v5.3
15	Amenities	Play Area	SC_10_25_40	DM_GIS_BDDS_v5.3
16	Amenities	Swimming Pool	SC_10_25_45	DM_GIS_BDDS_v5.3
17	Amenities	Swimming Pool Deck	SC_10_25_50	DM_GIS_BDDS_v5.3
18	Amenities	Sport Facility	SC_10_25_55	DM_GIS_BDDS_v5.3
19	Amenities	Prayer Room	SC_10_30_10	DM_GIS_BDDS_v5.3
20	Amenities	Female Prayer Room	SC_10_30_15	DM_GIS_BDDS_v5.3
21	Amenities	Male Prayer Room	SC_10_30_20	DM_GIS_BDDS_v5.3
22	Amenities	Cinema	SC_10_35_10	DM_GIS_BDDS_v5.3
23	Amenities	Entertainment Area	SC_10_35_15	DM_GIS_BDDS_v5.3
24	Amenities	Staff Rest Area	SC_10_40_10	DM_GIS_BDDS_v5.3
25	Amenities	Waiting Area	SC_10_40_15	DM_GIS_BDDS_v5.3
26	Amenities	Landscape	SC_10_40_20	DM_GIS_BDDS_v5.3
27	Amenities	Multipurpose Hall	SC_10_45_10	DM_GIS_BDDS_v5.3
28	Amenities	Clinic	SC_10_50_10	DM_GIS_BDDS_v5.3
29	Amenities	Ironing Room	SC_10_55_10	DM_GIS_BDDS_v5.3
30	Amenities	Spa	SC_10_60_10	DM_GIS_BDDS_v5.3
31	Circulation Areas	Entry Hall	SC_15_10_10	DM_GIS_BDDS_v5.3

Dubai BIM Standard & openBIM

- ❑ The Standard defines the information requirements for building permit submissions.
- ❑ The Industry Foundation Classes (IFC) format is how the information (BIM models) is submitted/delivered.

Why openBIM IFC?

Open

Open standards facilitate industry interoperability

Reliable

Data exchanges depend on reliable, independent quality benchmarks

Flexible

Choose the right technology for you, creating more value to all stakeholders

To know more about open BIM standards:



<https://www.buildingsmart.org>



<https://buildingsmartuae.ae>

Dubai BIM Standard – Download and Updates

- ❑ The latest Dubai BIM Standard files and samples from the Build in Dubai website:
<https://buildindubai.gov.ae/bim>
- ❑ The current version is 1.4.
- ❑ The standard is subject to future updates.
- ❑ so, it is highly recommended to download the latest version before starting a new submission.

Technical Guidance

Key Considerations for BIM Submissions



DM BIM Templates



Modelling Recommended Practices



Minimum Information Requirements



Model Geolocation



Model Segregation



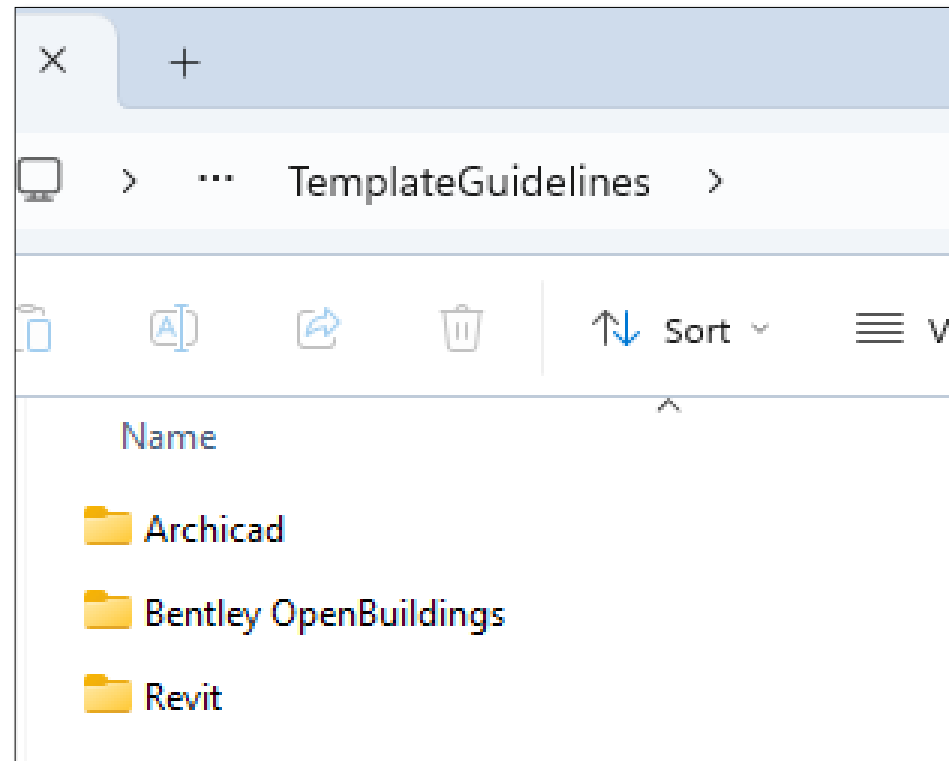
Model Export



Simple Self-Checks
before Submission

Technical Guidance - DM BIM Templates

- ❑ DM BIM Templates is included in the downloadable DM BIM standard
- ❑ DM BIM Templates are recommended to use but not mandatory



Technical Guidance - Model Geo-referencing

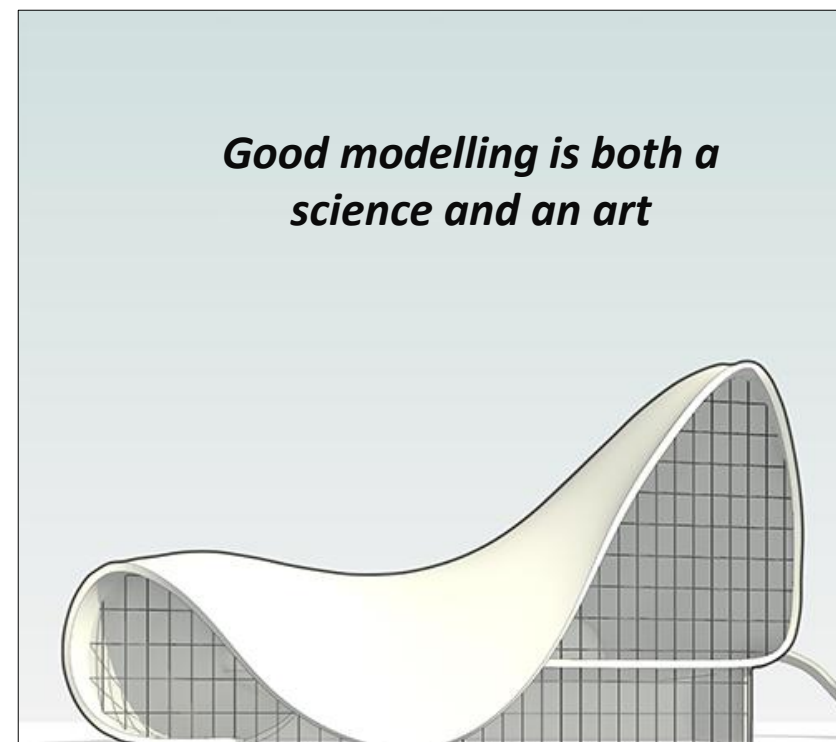
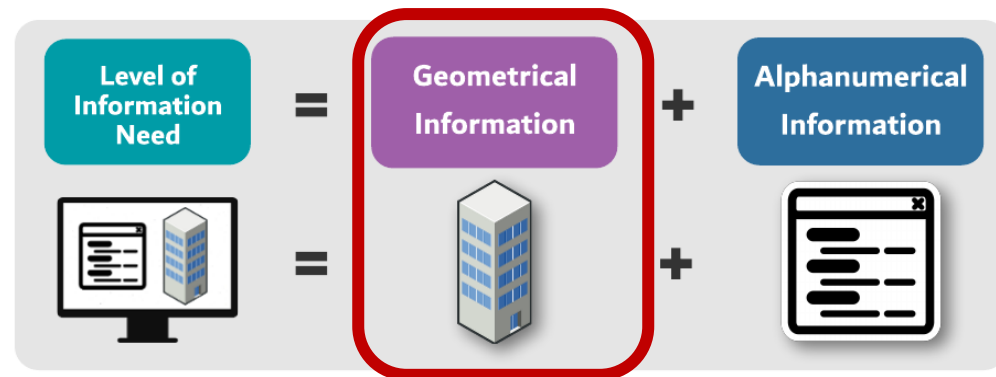
Uploaded models to the BIM E-Submission platform must be georeferenced

- ☐ It is recommended to geo-reference the model from the start of the modelling process to avoid major change later.
- ☐ Models can be uploaded the E-BIM submission platform anytime to check geo-location.
- ☐ Download the land parcel information using the GIS service on Build in Dubai website and use DXF file for the parcel details.
- ☐ Check the supporting documents under the standard for help on Geo-referencing.

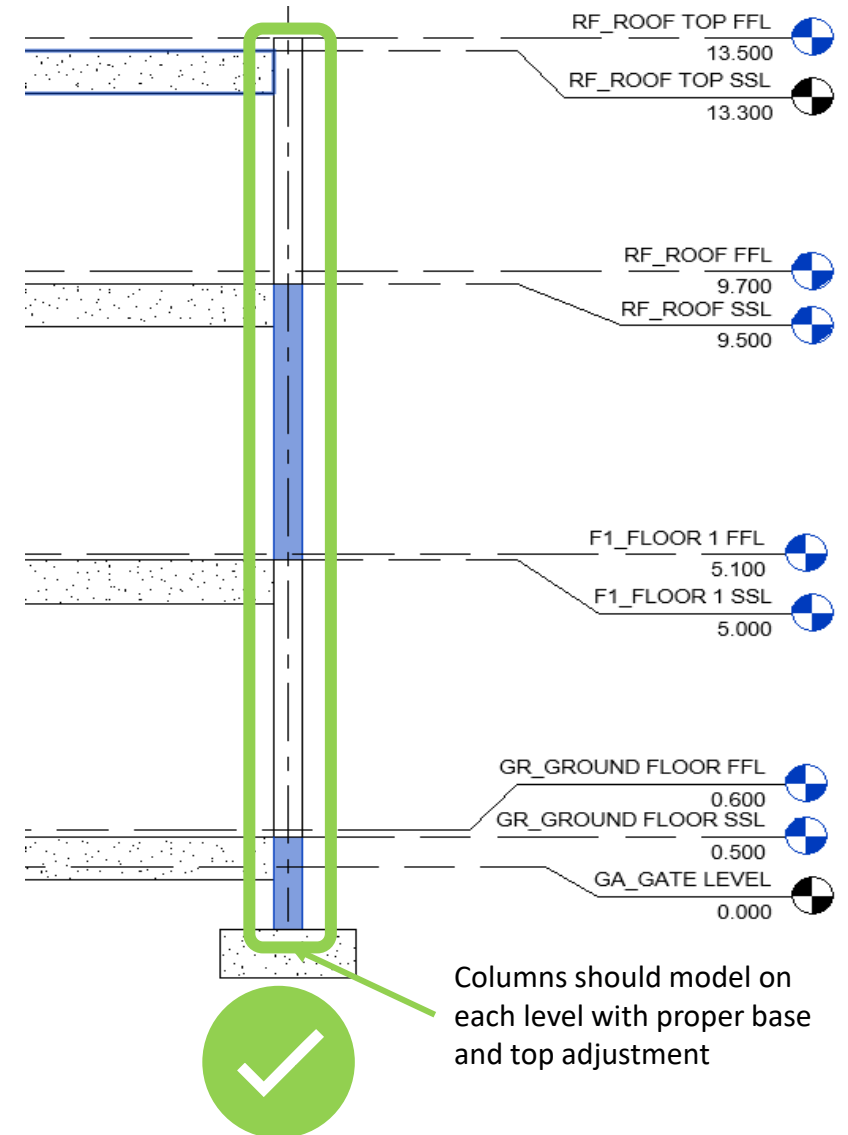
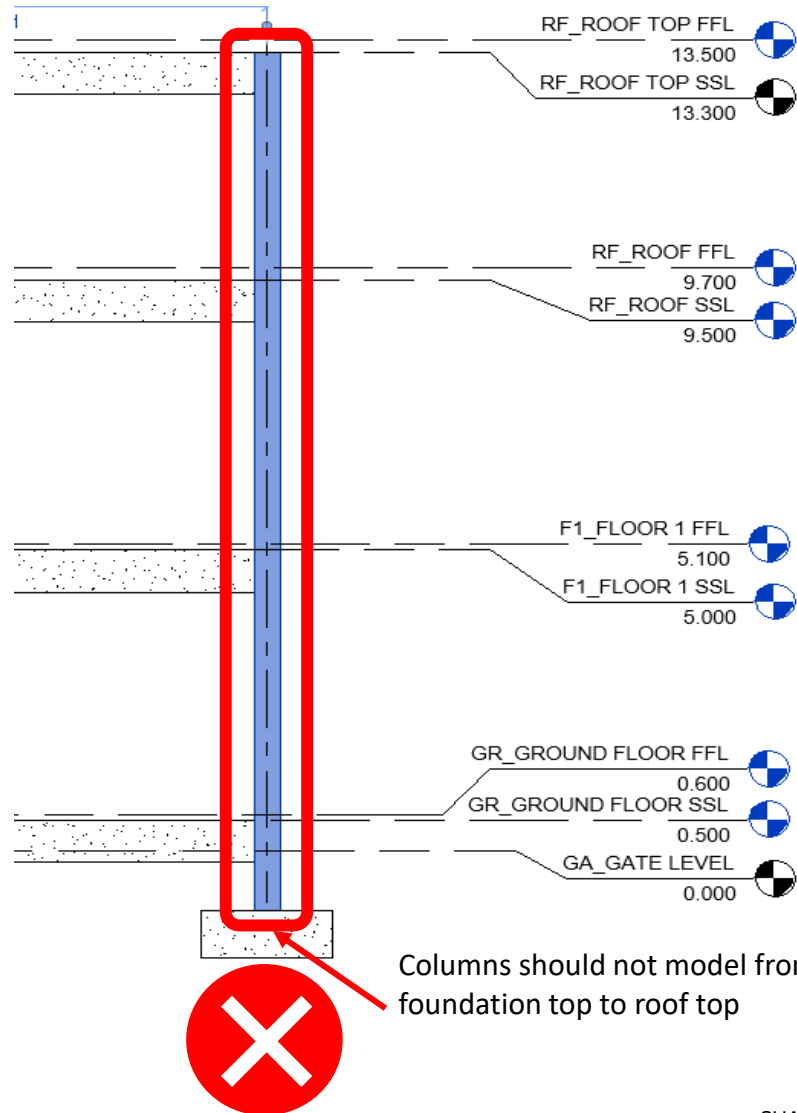


Technical Guidance - Modelling Recommended Practices

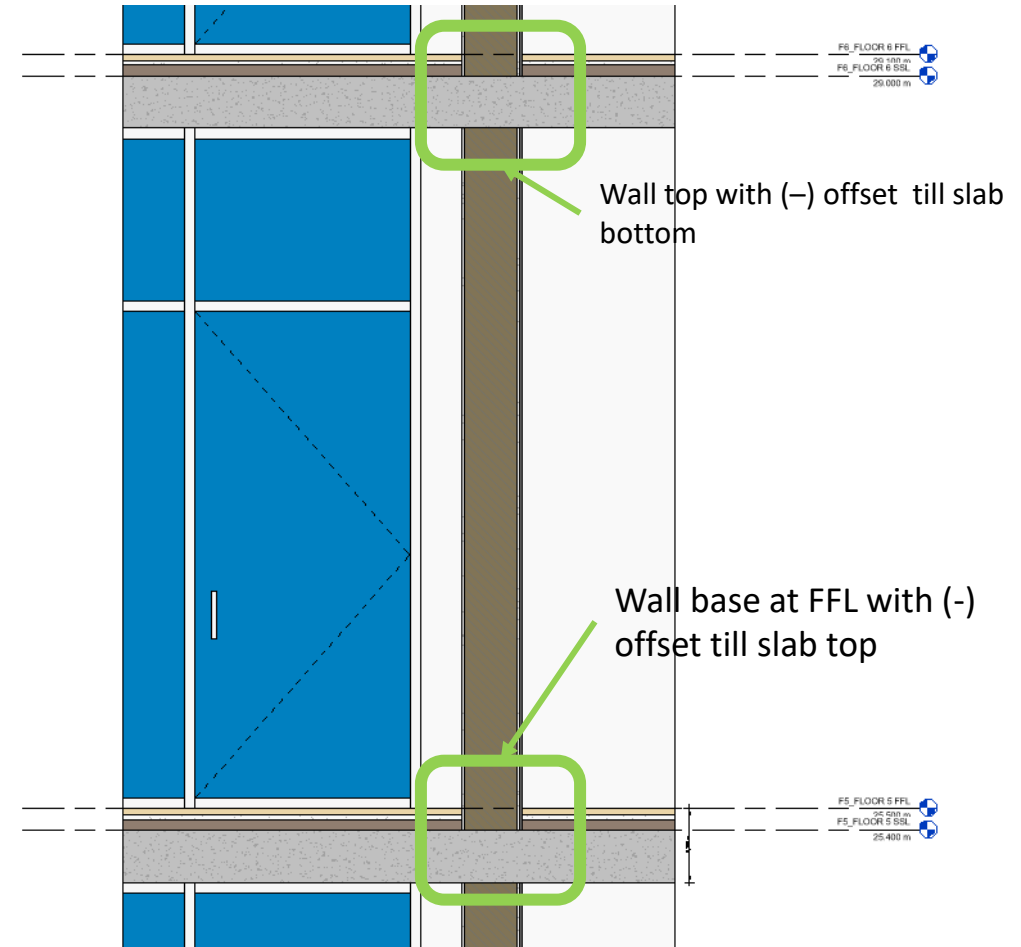
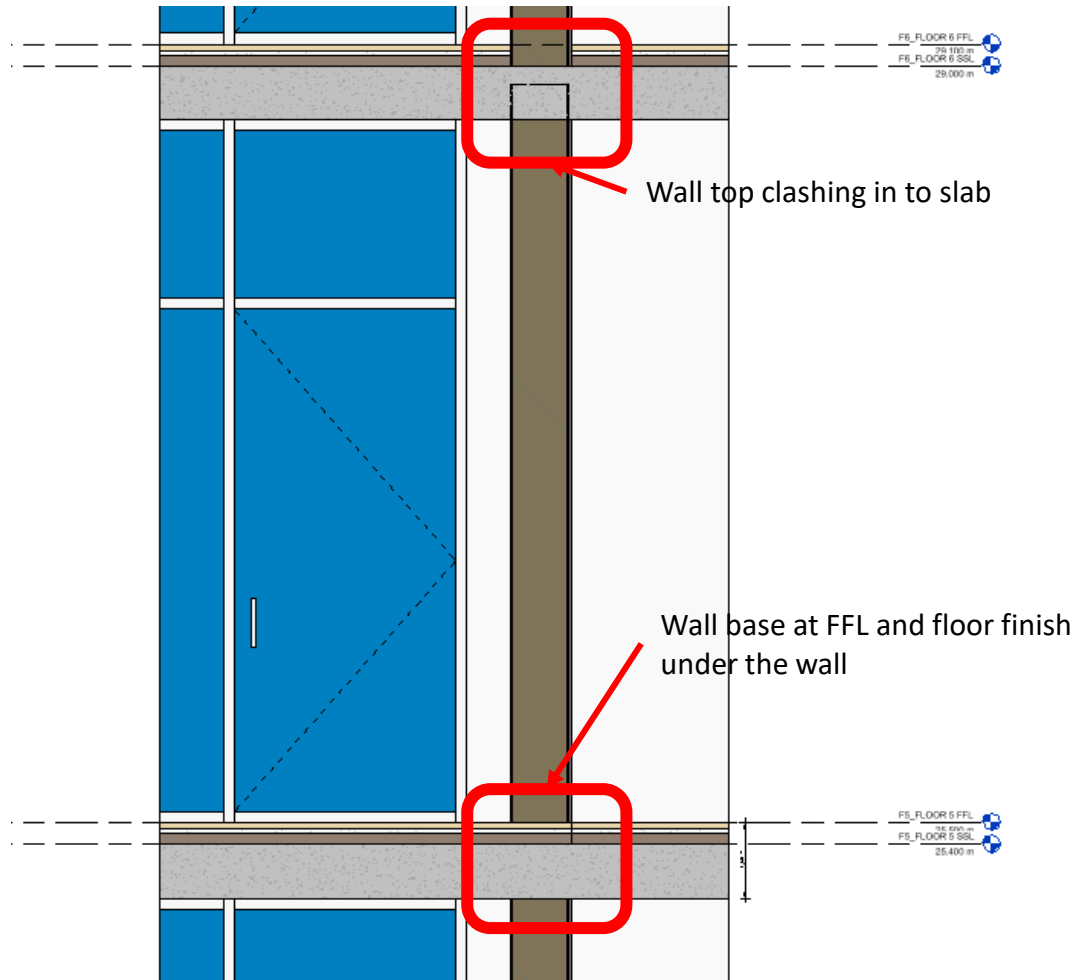
- ❑ Certain modelling styles are not acceptable for DM BIM submission.
- ❑ Modelling for DM BIM submission is by large based on universal modelling best practices.
- ❑ Adopting the recommended modelling practices:
 - ✓ Ensures better model visualization experience
 - ✓ Allows automatic validation and e-checking of the model
 - ✓ Assists with the manual review of the models by DM BIM & permit engineers
- ❑ For the full list, check the Modelling Recommended Practices On <https://buildindubai.gov.ae/bim>
- ❑ Download and check the samples to check how model geometry is created



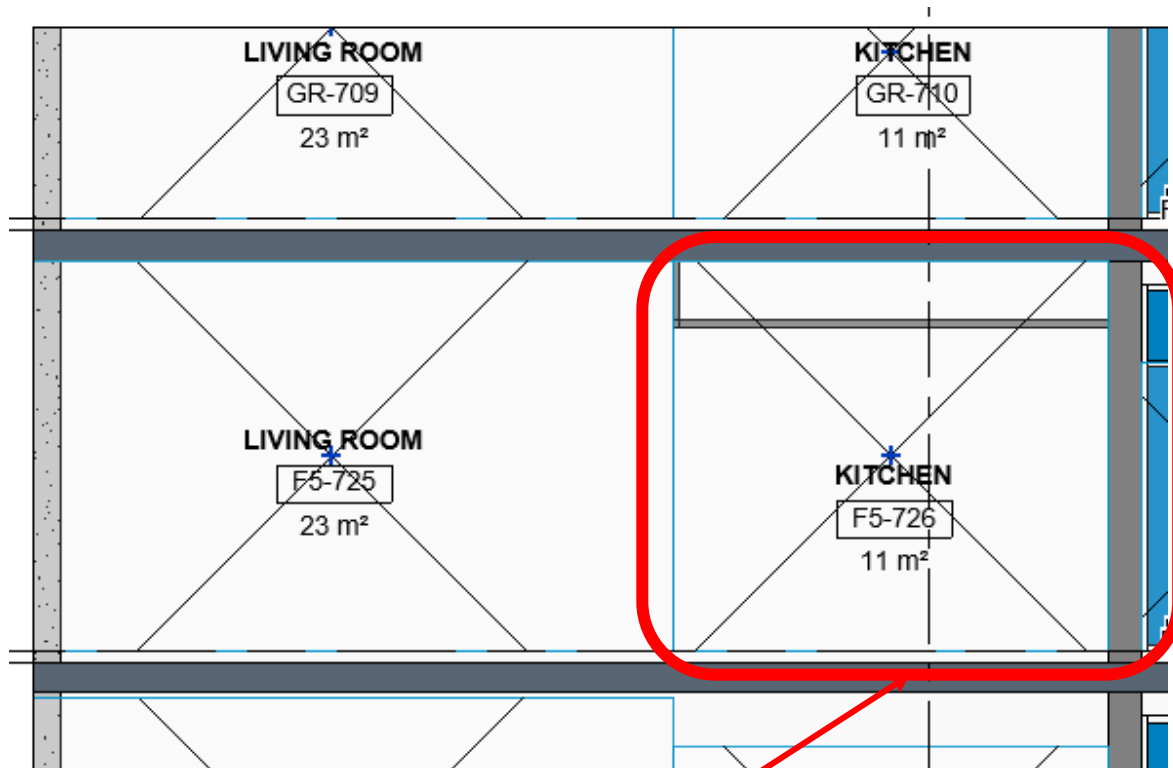
All columns should be modeled in its corresponding level



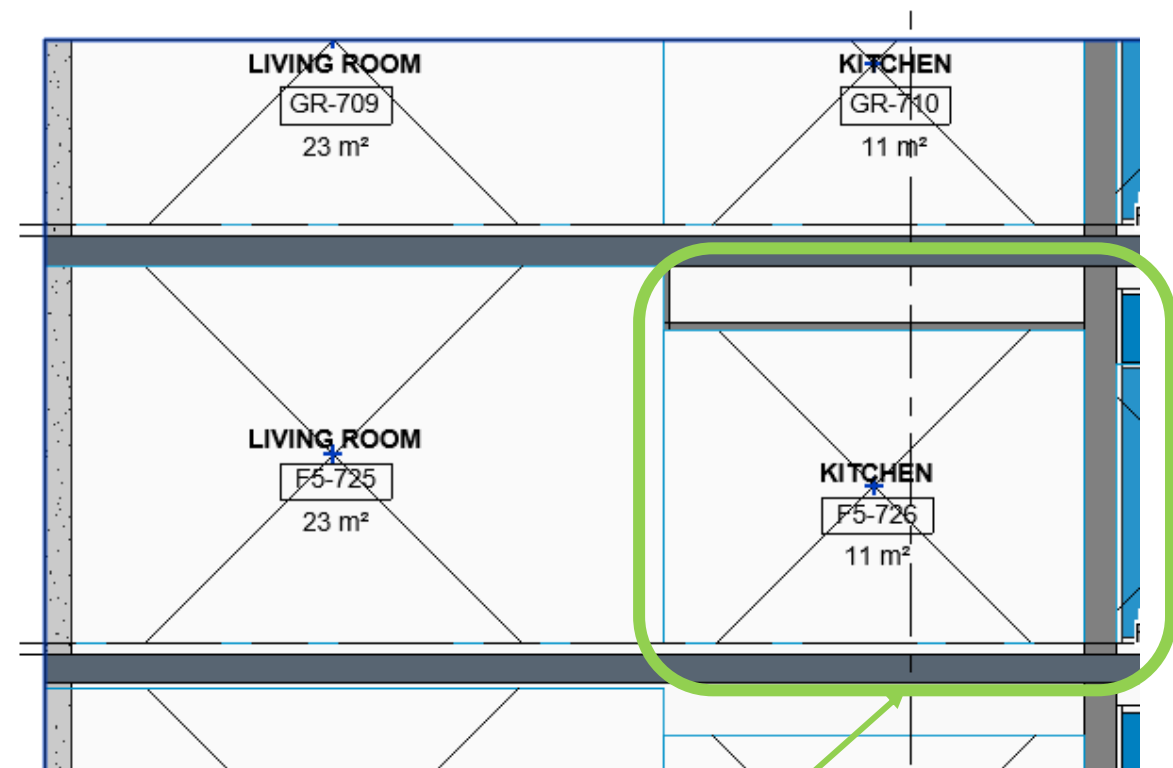
Wall should be modeled from FFL level with base (-) offset till SSL level and top should be till the slab bottom



Space height should be adjusted



Spaces height till slab bottom crossing ceiling .



Space height adjusted till the ceiling.

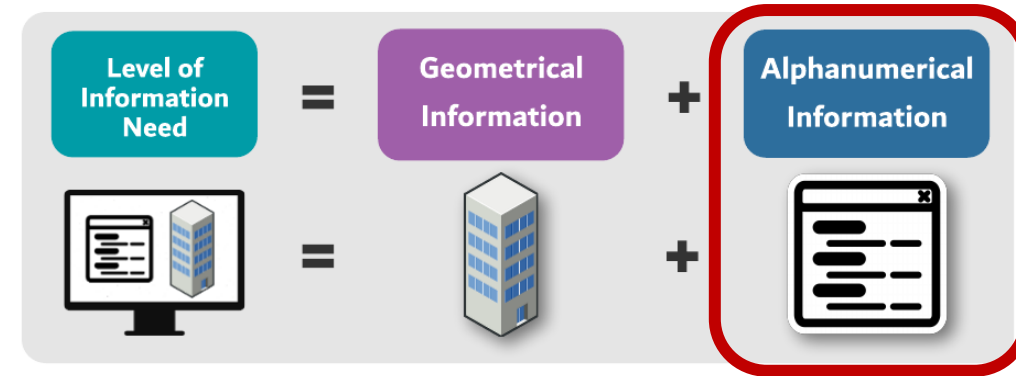


Technical Guidance - Minimum Information Requirements

❑ What Is Required?

- ✓ The required Information defined in **Appendix B – Model Attribute Matrix**
- ✓ Information must be added to key model elements (Spaces, Slabs & Walls, etc.).

(BIM Model > 3D Model)



❑ Why? The model information enriches the model to support:

- ✓ Auto-generating the building card (e.g. BUA, total parking slots, etc.)
- ✓ Auto checking some rules according DBC (e.g. area and dimension rules)
- ✓ Validation against submitted 2D drawings and manual inputs on the system

❑ Download and check the samples to learn how and where the information is added

Technical Guidance - Model Segregation

A project may include multiple buildings with each building made of multiple parts

Model segregation is allowed, but keep in mind:

1. All IFC files (AR and ST) should be submitted in a single zip file.
2. No mix of AR and ST elements in any file.
3. No elements duplicates between files.
4. Each IFC file should represent a whole building or part of building.
5. Claddings/landscape of a building can be stored in separate files.
6. IFC files should be named correctly as per the standard file name convention.



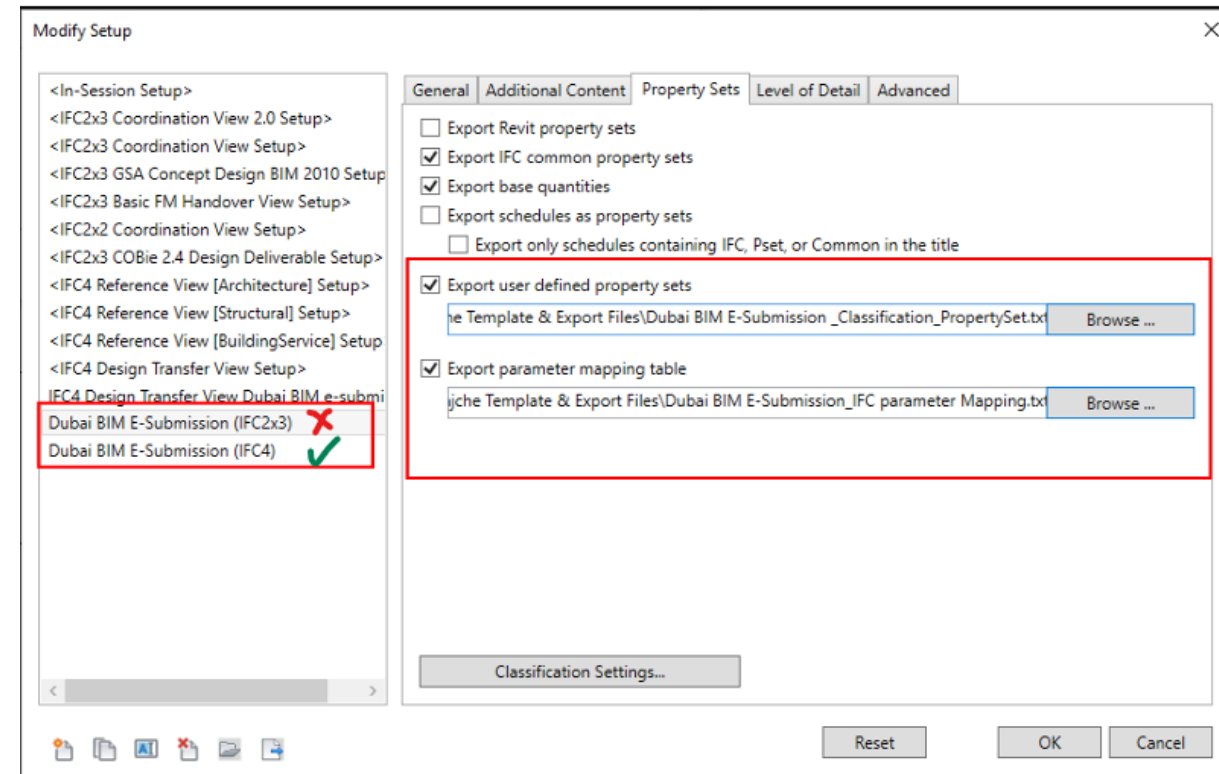
DubaiBIMTechnicalGuides.pdf provides clear details on model segregation

Technical Guidance - Model IFC Export

IFC Export should be completed correctly to ensure compliance with submission requirements.

Things to remember:

- ❑ BIM models need to be export as **IFC4**
- ❑ Many problems with exported IFC files such as bad visualization & large files are due to incorrect exporting.



Technical Guidance - Self-Check

Exported IFC files

- PN299465_BI000001_PA2320237_AR_001_V4.ifc
- PN299465_BI000001_PA2320237_ST_001_V4.ifc
- PN299465_BI000002_PA2320237_AR_001_V4.ifc
- PN299465_BI000002_PA2320237_ST_001_V4.ifc
- PN299465_BI000003_PA2320237_AR_001_V4.ifc
- PN299465_BI000003_PA2320237_ST_001_V4.ifc

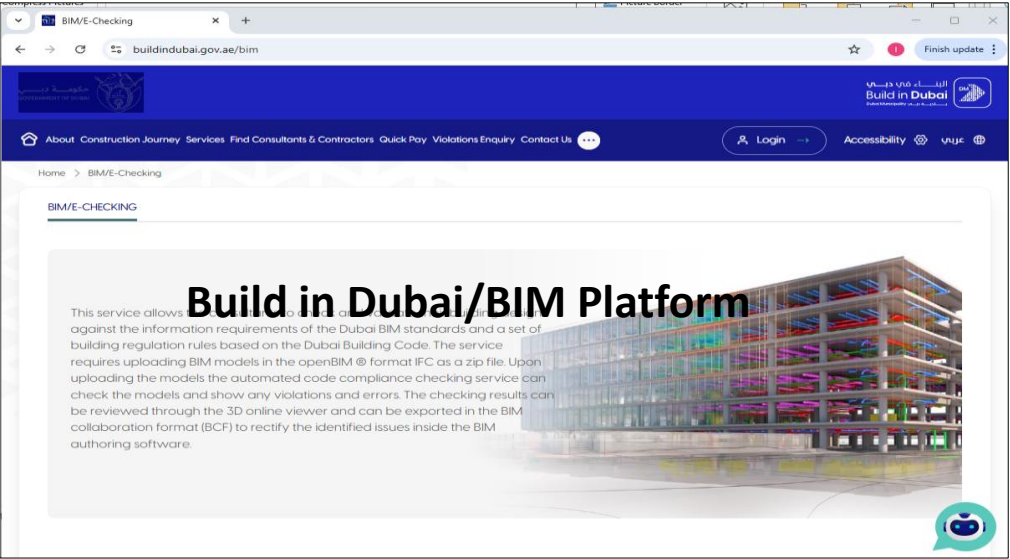
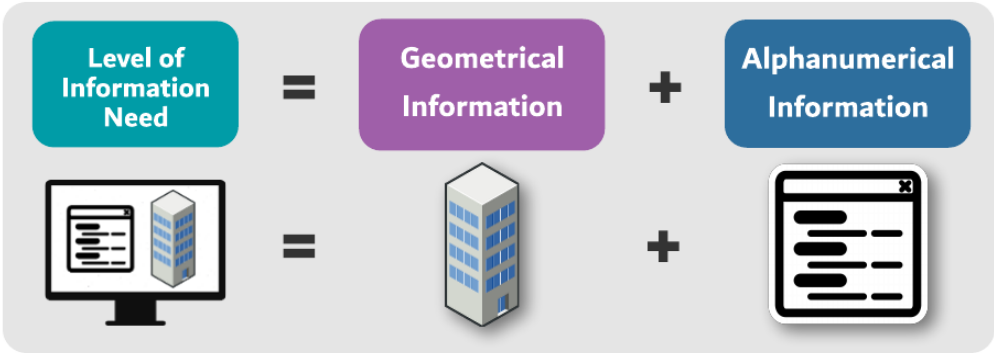
PN299465_BI000001_PA2320237.zip

Local/ Offline

Online

Desktop IFC Viewer

DM QA/QC APPs



Wrapping up...

Getting ready for DM BIM Submission? your team needs:

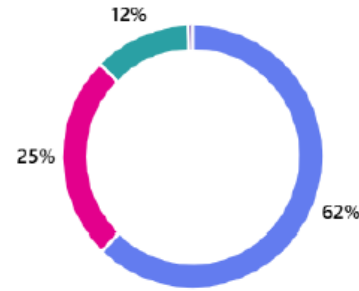
- 1) **Understand** the mandate scope and BIM permitting process
- 2) **Download and review** DM BIM Standard, samples, supporting documents, recorded workshops.
- 3) **Build** some knowledge of IFC (Industry Foundation Classes) and exporting models from your BIM authoring tool
- 4) **Model** with DM submission requirements and modeling best practices in mind
- 5) **Find out** how to self-check submissions locally using DM Apps and free IFC Viewers
- 6) **Find out** how to self-check submissions on the DM BIM online platform, including the Automated Code Compliance results and geo-referencing.
- 7) **Submit** to DM engineers for review.

Insights from recent questionnaire on BIM adoption in Dubai

استخدام برامج ال CAD او ال BIM لتجهيز المخططات الهندسية للحصول على رخصة البناء

Do you use BIM to produce 2D CAD drawings for permit submission?

Yes, all CAD drawings are extracted from BIM models	109
No, we still using CAD tools to generate the 2D drawings from scratch.	44
Not Applicable	21
Other	1

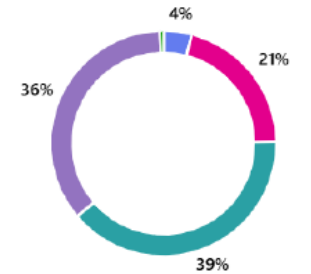


- 71% من الشركات تستخدم ال BIM لإنشاء المخططات
- 29% من الشركات مازالت تستخدم ال CAD بشكل أساسي لإنشاء المخططات

مستوى تطبيق نمذجة معلومات البناء في الشركات

How would you rate your company's BIM expertise?

None	7
Basic	36
Moderate	68
Advanced	62
Other	1



- 39% يعتبرون ان شركاتهم تستخدم نمذجة معلومات البناء بشكل متوسط
- 36% يعتبرون ان شركاتهم تستخدم نمذجة معلومات البناء بشكل متقدم
- 21% يعتبرون ان شركاتهم تستخدم نمذجة معلومات البناء بشكل بسيط
- 4% فقط من الشركات لا تستخدم نمذجة معلومات البناء

Introduction to IFC

Do you like getting locked inside proprietary file formats, or would you prefer the freedom of Open BIM?

IFC

Industry Foundation Classes — the open, vendor-neutral data format that lets BIM software speak the same language.



What is IFC?

Industry Foundation Classes (IFC) is an **open, neutral, vendor-independent** data format for describing building and construction industry data.

- Developed and maintained by buildingSMART International
- Free to use — not owned or controlled by any single software vendor
- Enables "openBIM" — collaborative workflows across any compliant tool
- Covers geometry, materials, relationships, and project data

ISO 16739

the official international
standard for IFC

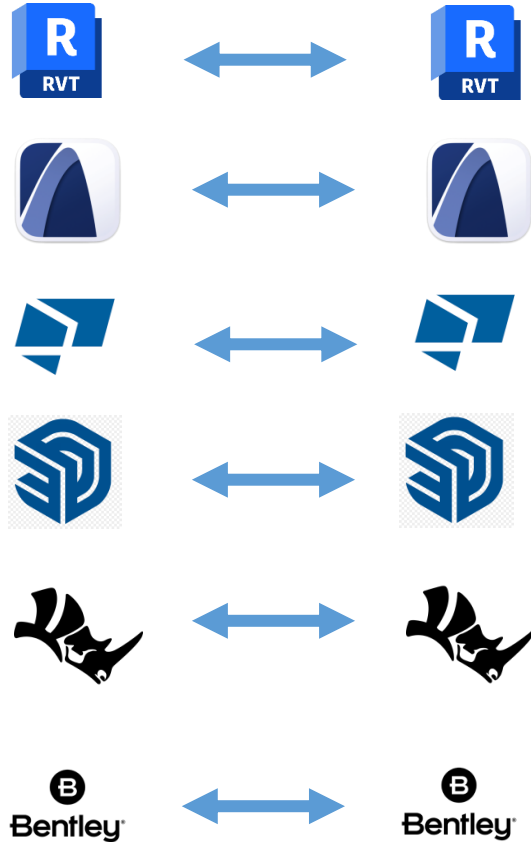


buildingSMART

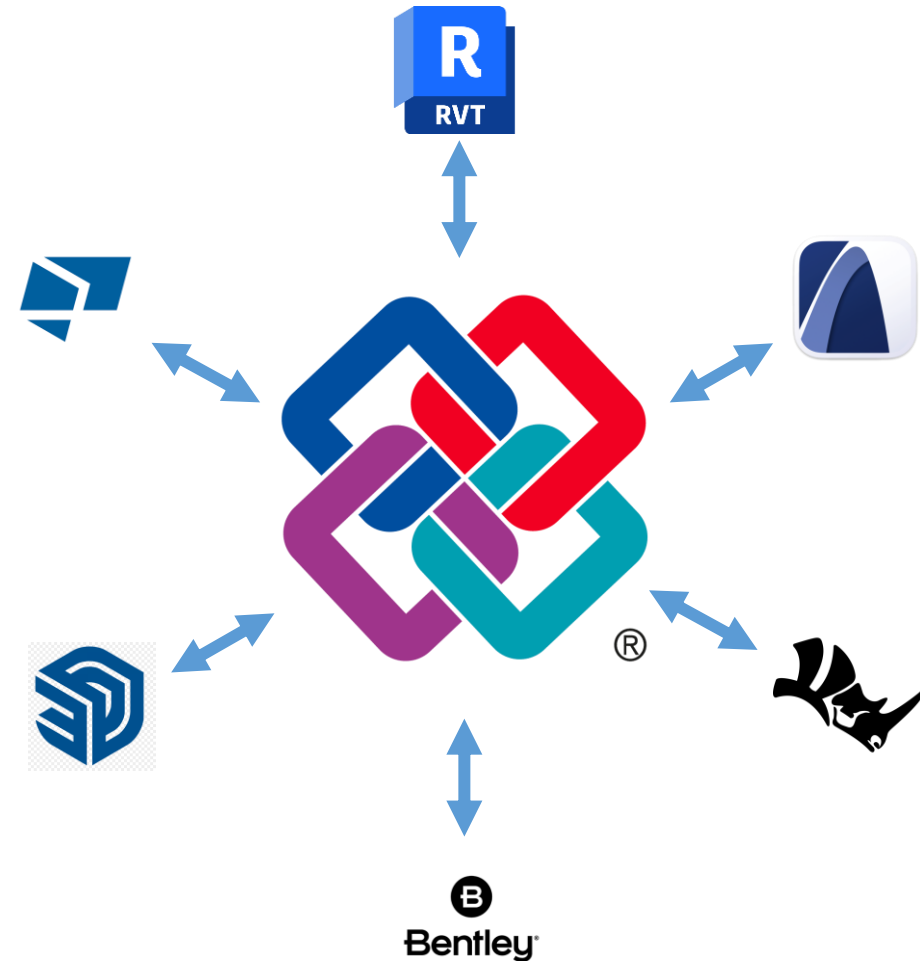
International non-profit that authors and maintains the IFC schema as an open standard for the whole industry.

The BIM Interoperability

Closed BIM



Open BIM



BIM interoperability with IFC allows different BIM software applications to exchange and understand building information using a common, open standard, enabling efficient collaboration throughout the design, construction, and operation of a building.



How IFC Works

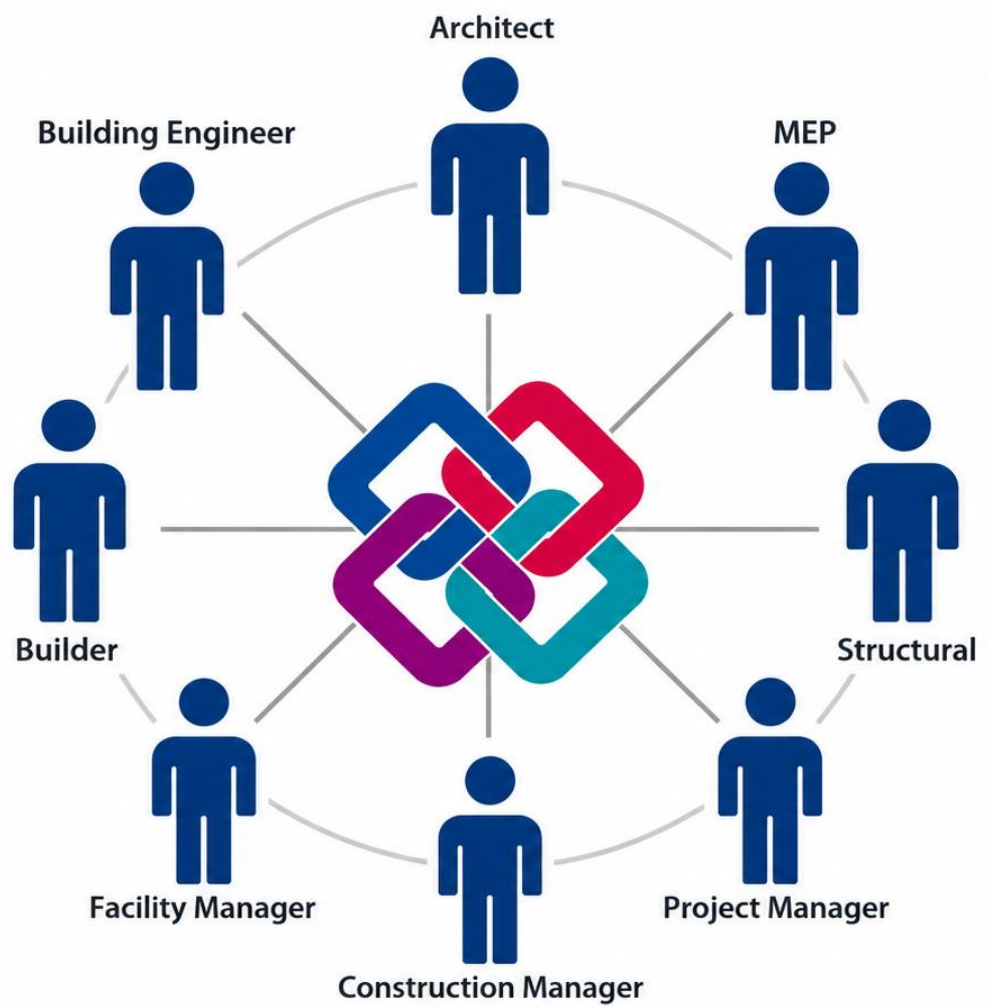
IFC acts as a neutral hub between specialist software

One open format, many tools

Same .ifc file in and out

No proprietary lock-in

Round-trip collaboration





Inside an IFC File

A structured, text-based hierarchy of building data

- IFC files (.ifc) are plain text, following the STEP / EXPRESS schema
- Variants: .ifc (text), .ifcXML, .ifcZIP (compressed)
- Every element is an entity with properties and relationships





IFC in Practice

Common real-world use cases



Clash Detection

Spot conflicts between disciplines before construction begins



BIM Coordination

Multi-disciplinary teams review and merge models in shared viewers



Facility Management

Handover rich building data to owners and operators



Quantity Takeoff & Cost

Extract quantities and properties for accurate estimating

Popular open viewers: BIMvision · Solibri Anywhere · usBIM Viewer · BIMcollab Zoom